

PilotFish EDI Transformation Processor – Complete Reference (26R1.11 WAR)

Derived from decompiled eip.war.hs.26R1.11

August 4, 2026

At a glance (plain English)

Think of this as a **translator between EDI and XML**. Raw X12 text becomes structured EDI XML for mapping, validation, and routing — or XML is rendered back into EDI segments using your delimiter and table-data settings.

EDI Transformation — what it does.

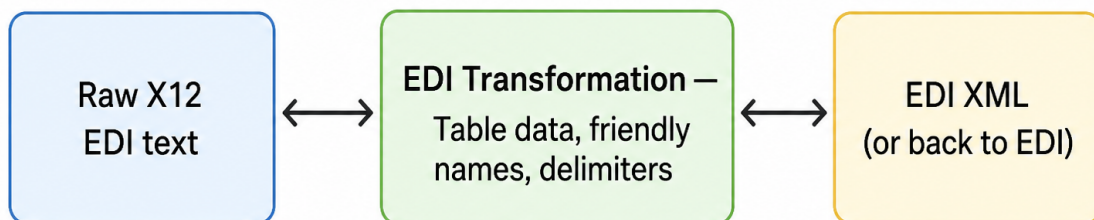


Figure 1: At a glance (plain English)

- **Converts** raw X12 EDI to **XML** (default direction) or **XML to EDI**.
- **Uses** X12 table data for loops, friendly names, and code definitions.
- **Writes** discovered delimiters to transaction attributes on EDI-to-XML.
- **Hides** irrelevant configuration tabs based on transformation direction.

Purpose

This document is a **deep dive** into the **EDI Transformation** processor as shipped in **eip.war.hs.26R1.11**. The processor class delegates to **EDITransformationModule** via **AbstractTransformationProcessor**.

Provenance	Value
WAR	eip.war.hs.26R1.11
Module JAR	xcs-format-edl-1.0.10.jar
Core JAR	xcs-core-1.2.6.jar
Decompiler	CFR 0.152
Implementation class	com.pilotfish.eip.modules.transform.EDITra
Module implementation	com.pilotfish.eip.modules.transform.edi.ED
Decompiled path	war-pipeline/decompiled/eip-26R1.11/.../ED
Processor type string	Transformer - EDI / XML
Description	Transforms EDI to / from XML
Categories	HEALTHCARE, TRANSFORMATION

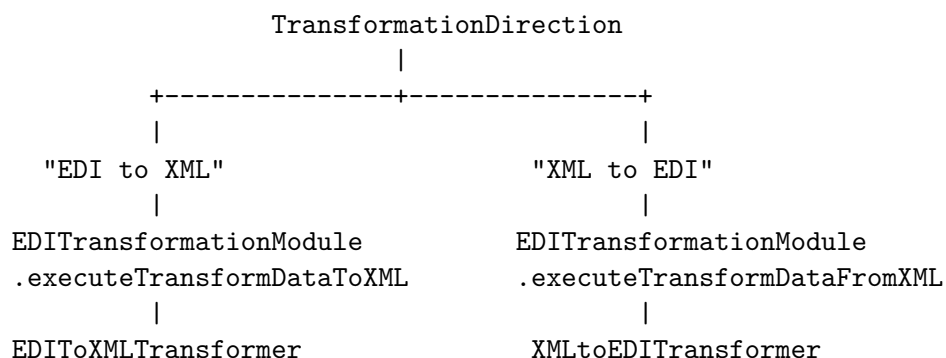
Related EDI stack (same JAR):

Class	Role
EDIToXMLTransformer	EDI text -> XML
XMLtoEDITransformer	XML -> EDI text
EdiIdentificationProcessor	Envelope attributes on raw EDI (separate processor)
EdiSNIPValidationProcessor	SNIP validation on EDI XML

Dialect guide: PilotFish-EDI-XML-Guide-26R1.11.md — document shape, modern vs legacy tags, XSLT patterns, and XML→EDI quirks.

Derived from WAR bytecode (CFR).

1. Conceptual model



+ table data / friendly names	+ delimiter config
EDI XML on cache stream	Raw EDI on cache stream
+ delimiter attributes	(no delimiter attrs written)

Supported direction: **TransformDirection.BIDIRECTIONAL**.

2. High-level behavior (**AbstractTransformationProcessor.processData**)

1. Read **TransformDirection** (TAG_DIRECTION).
2. Validate against allowed options (EDI to XML and/or XML to EDI).
3. Invalid direction -> **EIPEException**("Specified transformation direction \"...\" is not valid").
4. **XML to EDI** -> **EDITransformationModule.executeTransformDataFromXML(data)**.
5. **EDI to XML** -> **EDITransformationModule.executeTransformDataToXML(data)**.

Lazy-init: **initializeTransformer()** creates **EDITransformationModule** on first use; resources updated when loader/manager change.

3. EDI to XML algorithm (**executeTransformDataToXML**)

1. Read configuration flags: friendly names level, legacy format, include not-defined elements, code defs, enhanced context, segment indexing.
2. **buildEdiDataVersions(data)** when enhanced context enabled — loads X12 table data directories (4010/5010) or bundled trial data.
3. Construct **EDIToXMLTransformer** with table versions and options.
4. Copy EDI stream through transformer into **EIPCacheManager default cache**.
5. Set delimiter **attributes** on the transaction:

Attribute key	Source
<code>com.pilotfish.eip.modules.transform.edi.EDITransformationModule.transactionDelimiter</code>	<code>SegmentRecordDelimiter</code>
<code>com.pilotfish.eip.modules.transform.edi.EDITransformationModule.transactionElementDelimiter</code>	<code>ElementDelimiter</code>
<code>com.pilotfish.eip.modules.transform.edi.EDITransformationModule.transactionSubElementDelimiter</code>	<code>SubElementDelimiter</code>
<code>com.pilotfish.eip.modules.transform.edi.EDITransformationModule.transactionRepetitionDelimiter</code>	<code>RepetitionDelimiter</code>
	<code>(\u0000)</code>

6. `data.setDataStream(cache.getInputStream());` return data.
7. Failure -> **EIPEException**("Error transforming EDI to XML", cause).

XML shape: Modern path uses `/interchange/group/transaction`; legacy mode uses `/ediroot/interchange/group/transaction` and `LOOP_` prefixes.

4. XML to EDI algorithm (`executeTransformDataFromXML`)

1. Borrow **DocumentBuilder** from module pool (secure processing, namespace-aware).
2. Parse XML from transaction stream.
3. Construct **XMLtoEDITransformer** with:
 - `OutputEndlines`, `DataDelimiter`, `SubDelimiter`, `RepetitionDelimiter`, `RecordDelimiter`, `UseProvidedDelimiters`
4. Write EDI to cache via **OutputStreamWriter** on cache output stream.
5. `data.setDataStream(cache.getInputStream());` return pooled document builder.
6. Failure -> `EIPEException("Exception processing XML to EDI", cause)`.

When **UseProvidedDelimiters** is true, delimiter attributes on the XML interchange element override configured defaults when present.

5. Configuration reference

Configuration merges **AbstractProcessor**, **AbstractTransformationProcessor**, **AbstractTransformationModule** and **EDITransformationModule**.

5.1 Transformation direction

UI label	Tag	Type	Default	Notes
Transformation	<code>TransformationDirection</code>	Multiple choice	first option (EDI to XML)	Required

Tab visibility dependencies:

Direction	Hidden tabs
EDI to XML	Transform to EDI tab
XML to EDI	Table Data, Enhancements, Compatibility

5.2 Conditional execution (two flags)

UI label	Tag	Default	Tab
Execute Processor	<code>ExecuteProcessor</code>	<code>true</code>	Conditional Execution
Execute Transformation	<code>ExecuteTransformation</code>	<code>true</code>	Conditional Execution

The transformation module's execute flag is embedded via `extendConfigurationDescriptor`; processor skip is evaluated by the framework before `processData`.

5.3 Table Data tab

UI label	Tag	Type	Default	Notes
Use X12 Table Data for Enhanced Functionality	USE_ENHANCED_CONTEXT	Boolean	true	Master switch for table-driven features
Use Bundled X12 Table Data	UseInternalData	Boolean	true	Shown only when trial/bundled data valid
Compatibility Table Data List	CUSTOM_METADATA_DIRECTORY	String	""	Legacy dirs assumed 5010
X12 Table Data Directories	TransactionDataWithVersion	Multiple choice (Directory / Version)	5010	File picker + version dropdown (4010, 5010)

Dependencies: When enhanced context is false, table/enhancement/compatibility items are not required for the to-XML path logic (tabs may still show until direction hides them).

Trial data: If bundled trial expired and no directories configured -> `EIException("Error! X12 Trial Table Data has expired...")`.

5.4 Enhancements tab

UI label	Tag	Type	Default
Add Code Definitions	CODE_DEFS	Boolean	false
Friendly Naming	FriendlyNamesLevel	Multiple choice	None
Add Segment Indexing	SegmentIndexing	Boolean	true

Friendly Naming choices: None, Add to element name, Add as attributes.

5.5 Transform to EDI tab (XML -> EDI)

UI label	Tag	Type	Default
Output Endlines	OutputEndlines	Boolean	true
Use Provided Delimiters if Available	UseProvidedDelimiters	Boolean	false
Element Delimiter	DataDelimiter	String	*
Sub Delimiter	SubDelimiter	String	:
Repetition Delimiter	RepetitionDelimiter	String	^
Record Delimiter	RecordDelimiter	String	~

5.6 Compatibility tab

UI label	Tag	Type	Default
Output legacy XML format	USE_LEGACY_FORMAT	boolean	false
Include Not Used/Defined Data Elements	INCLUDE_NOT_DEFINED_ELEMENTS	boolean	true

Legacy migration (readConfiguration): Routes missing `USE_ENHANCED_CONTEXT` get enhanced context **false**, legacy format **true**, and old boolean friendly/validation flags mapped forward.

6. X12 table data loading

For each directory/version pair:

1. Resolve absolute path (interface-relative via `ResourceLoader` when needed).
2. Read `sethead.txt` — comma-separated version map lines.
3. Build `EdiVersion` objects cached in `VERIFIED_DATA_DIRS` for reuse.
4. Missing/unreadable directory -> `EIException("Data directory: ... cannot be read or doesn't exist.")`.

Bundled trial path: `EdiDataCache.getTrialEdiVersions()` when license/trial window valid (90 days from license **Created** date logic in `checkForValidTrialData`).

7. Worked example — inbound 837

1. **EDI Identification** (optional) on raw EDI for envelope attributes.
2. **EDI Transformation**, direction **EDI to XML**, enhanced context **true**, friendly names **Add as attributes**, segment indexing **true**.
3. Output XML under `/interchange/group/transaction` with `SegIdx` attributes; delimiters on transaction attributes for later XML-to-EDI.

Reverse route: direction **XML to EDI**, **Use Provided Delimiters** true to honor attributes from step 2.

8. Known quirks and caveats

1. **UI type string** is `Transformer - EDI / XML`, not the shorter catalog name `EDITransformationProcessor`.
2. **Enhanced context strongly recommended** for healthcare maps — without table data, loops and code defs degrade.
3. **Trial table data expires** — production routes need explicit X12 table data directories.
4. **Saxon professional transformer pool** used for EDI-to-XML (`ProfessionalTransformerFactory`).
5. **Legacy XML** changes root paths and loop prefixes — mix legacy and modern processors carefully in one route.

6. **Include not-defined elements default true** — pre-26R1 routes without the tag get **false** via migration logic.
7. **Identification vs transform** — run identification on **raw EDI** before transform; SNIP validation expects **EDI XML** after transform.

9. Operational recommendations

Goal	Guidance
HIPAA routes	Enable enhanced context + appropriate 5010 table data
Mapping	Use friendly names mode that matches your map tool (attributes vs element names)
Round trip	Preserve delimiter attributes between to-XML and from-XML steps
Performance	Verified directory cache avoids re-parsing <code>sethead.txt</code>
Versioning	Match table data version to interchange implementation (4010 vs 5010)
Testing	Validate both directions in eiConsole with sample ISA/GS/ST payloads

10. Source index (this WAR)

Topic	Path under war-pipeline/decompiled/eip-26R1.11/
EDI Transformation processor	com/pilotfish/eip/modules/transform/EDITransformationProcessor
EDI transformation module	com/pilotfish/eip/modules/transform/edi/EDITransformationModule
AbstractTransformationProcessor	com/pilotfish/eip/extend/AbstractTransformationProcessor
EDI -> XML engine	com/pilotfish/eip/modules/transform/EDIToXMLTransformationEngine
XML -> EDI engine	com/pilotfish/eip/modules/transform/XMLtoEDITransformationEngine
Module JAR	war-pipeline/jars/eip-26R1.11/xcs-format-edi-1.0.10.jar

11. Pipeline provenance

PilotFish WARs/eip.war.hs.26R1.11

-> xcs-format-edi-1.0.10.jar

-> CFR decompile

-> Documents/Processors/26R1.11/PilotFish-EDI-Transformation-Reference-26R1.11.(md|pdf)

Deep-dive documentation from WAR decompile – EDI Transformation Processor – 26R1.11 – August 2026.